The conceptual origins of adjective ordering preferences: Evidence from silent gesture

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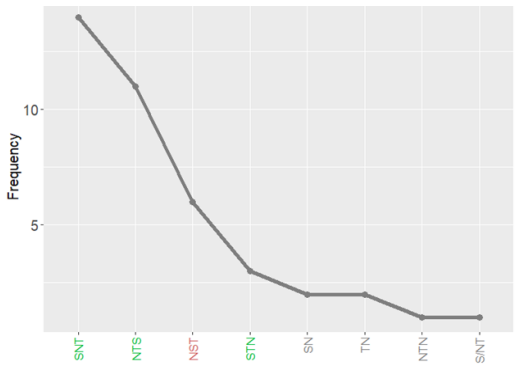
Languages display striking structural regularities despite considerable diversity. For example, it is widely suggested that, when multiple adjectives occur within a noun phrase, there is a typological tendency in terms of how these adjectives are ordered: languages that place adjectives before the noun tend to follow the same adjective order as English (e.g., *a big spotted toothbrush* sounds more natural than *a spotted big toothbrush*), while languages that place adjectives after the noun have been claimed to have an order that mirrors the English one (Dixon, 1982; Sproat & Shih, 1991). Syntactic (Cinque, 1994) and cognitive accounts (Sproat & Shih, 1991; Scontras et al., 2017, Culbertson et al., 2020) have been proposed to explain this typological tendency, but behavioral evidence supporting these accounts remains sparse.

In particular, one influential cognitive account proposes that adjective order reflects conceptual relationships: adjectives that are conceptually more closely related to the nouns they modify will appear linearly closer to the noun (Culbertson et al., 2020). For example, texture adjectives (e.g., *spotted*) are thought to be more closely associated with the objects they describe (e.g., *toothbrush*) than size adjectives (e.g., *big*), a relationship that can be captured using measures such as pointwise mutual information (Culbertson et al., 2020). However, it remains unclear whether such conceptual closeness shapes adjective ordering preferences independently of language-specific conventions.

Silent gesture has proved useful for bypassing the influence of native language word order (Goldin-Meadow et al., 2008). In addition, because it does not involve conventionalised syntactic concepts, it may reveal conceptual structures or biases that are relevant for explaining the ultimate origins of syntactic universals (Culbertson et al., 2020). In this study, we conducted two

silent gesture experiments to test whether ordering preferences among multiple adjectives conform to predictions based on conceptual closeness, in a modality unfamiliar to participants. In Experiment 1, we used a production task, where participants expressed, with gestures alone, pictures showing toothbrushes and pencils that varied in size and texture. Using logistic mixed-effects models, we observed the predicted preference only when size and texture adjectives were gestured prenominally: participants showed a preference for producing *size-texture-noun* orders over *texture-size-noun* orders ($\beta = 8.42$, $SE = 2.87$, $z = 2.94$, $p < 0.01$), similar to the English order (Figure 1). When size and texture were gestured postnominally, we only observed a non-significant trend towards producing *noun-texture-size* over *noun-size-texture* ($\beta = 0.01$, $SE = 1.76$, $z = 0.01$, $p = 0.99$) (Figure 1). In Experiment 2, we used a perception task, where participants saw the same set of pictures as Experiment 1 but chose between pre-recorded gesture sequences that were either homomorphic (i.e., following conceptual structure by placing texture adjectives closer to the noun than size adjectives) or non-homomorphic (i.e., placing size adjectives closer to the noun). Here we found no evidence of a preference for either type of order, regardless of whether the size and texture adjectives were prenominal ($\beta = 0.19$, $SE = 0.22$, $z = 0.85$, $p = 0.40$) or postnominal ($\beta = -0.17$, $SE = 0.22$, $z = -0.78$, $p = 0.43$) (Figure 2). Overall, the results do not provide clear support for a role of conceptual closeness in explaining cross-linguistic adjective ordering preferences, beyond the pre-nominal order found in English noun phrases. Future silent gesture studies examining a wider set of adjective classes with speakers of postnominal languages could further reveal the interplay between conceptual biases and native language influences.



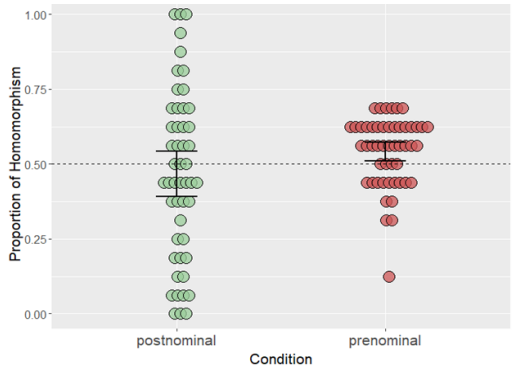


Figure 1 (left). The orders that were used in the majority of a participant's gesture productions. The x-axis shows the orders that at least one participant used in the majority of trials, and the y-axis shows

the number of participants who used each order in the majority of trials. S: size; N: noun; T: texture. When both modifiers were prenominal (STN/TSN), STN was preferred over TSN, as predicted; when they were both postnominal (NTS/NST), more NTS orders were produced than NST orders, but this preference was not statistically significant.

Figure 2 (right). The proportion of homomorphic orders selected by each participant. Each point represents a participant, the x-axis shows the conditions, and the y-axis shows the proportion of homomorphic orders selected. Participants did not show a preference for homomorphism in either the prenominal (STN, compared to TSN; in red) or postnominal (NTS, compared to NST; in green) condition.

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